

DDA3020 Machine Learning: L10 Convolutional Neural Networks

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03/25/2025

1 History of CNN

2 Convolutional Neural Networks

- Preview
- Convolutional layer
- Pooling layer
- Fully connected layer

1 History of CNN

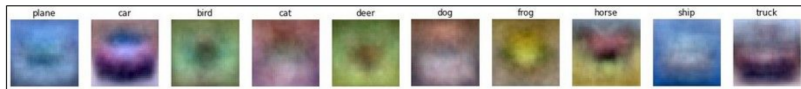
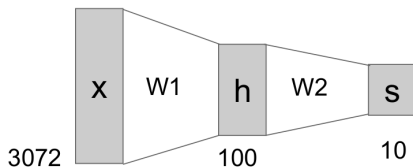
2 Convolutional Neural Networks

- Preview
- Convolutional layer
- Pooling layer
- Fully connected layer

Last week: Neural Networks

Linear model: $f(\mathbf{x}) = \mathbf{W}\mathbf{x}$

Two-layer neural network: $f(\mathbf{x}) = \mathbf{W}_2 \max(0, \mathbf{W}_1 \mathbf{x})$



Today: Convolutional Neural Networks

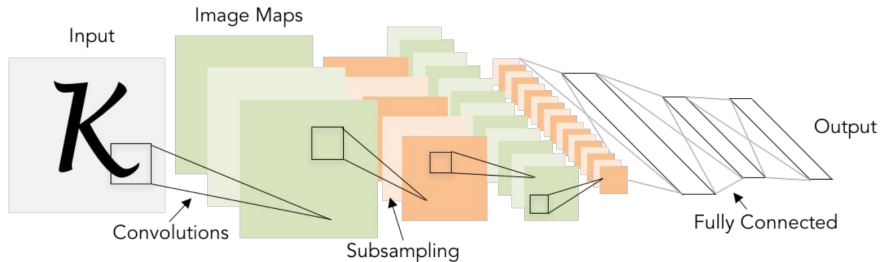
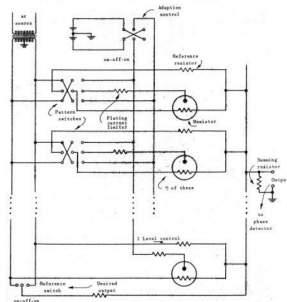
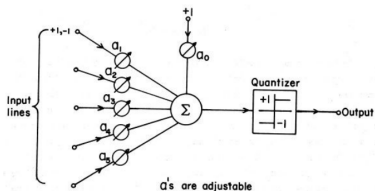


Illustration of LeCun et al. 1998 from CS231n 2017 Lecture 1

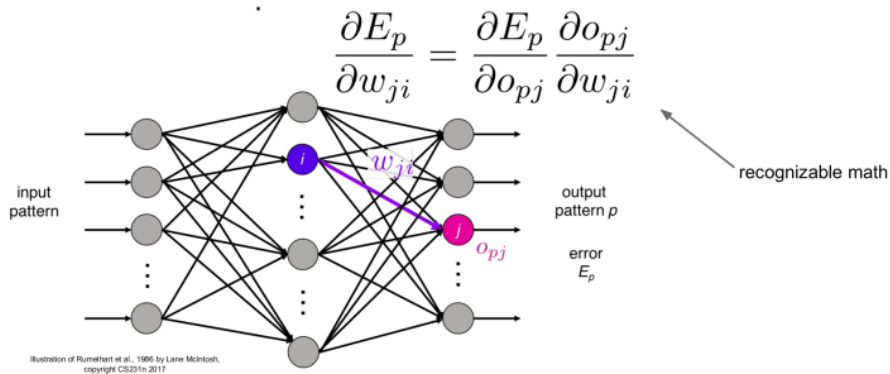
A bit of history...



Widrow and Hoff, ~1960: Adaline/Madaline

These figures are reproduced from [Widrow 1960, Stanford Electronics Laboratories Technical Report](#) with permission from [Stanford University Special Collections](#).

A bit of history...

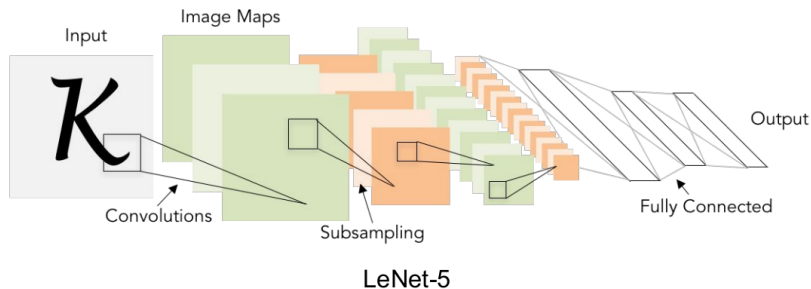


Rumelhart et al., 1986: First time back-propagation became popular

A bit of history:

Gradient-based learning applied to document recognition

[LeCun, Bottou, Bengio, Haffner 1998]



A bit of history...

[Hinton and Salakhutdinov 2006]
Reinvigorated research in Deep
Learning

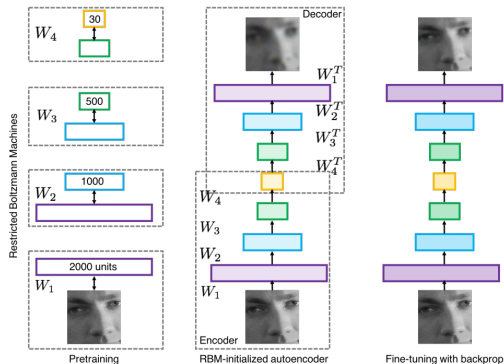


Illustration of Hinton and Salakhutdinov 2006 by Lane McIntosh, copyright CS231n 2017

A bit of history:

ImageNet Classification with Deep Convolutional Neural Networks [Krizhevsky, Sutskever, Hinton, 2012]

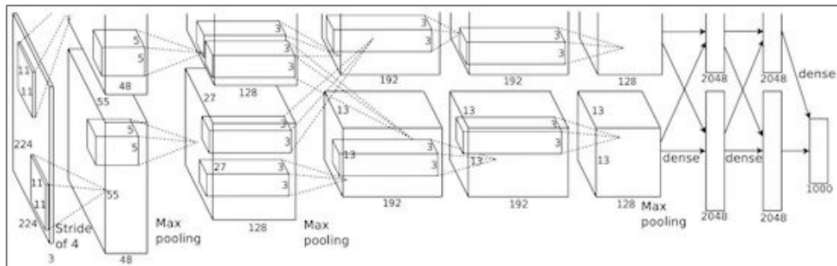
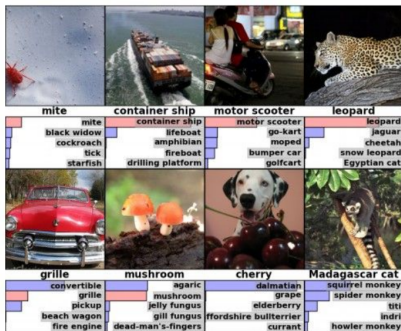


Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

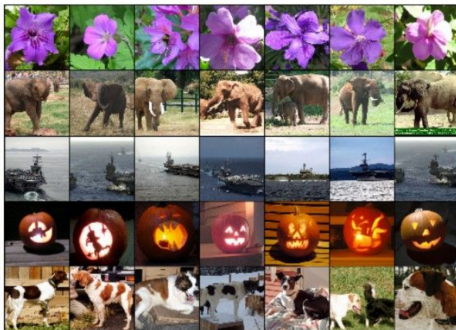
“AlexNet”

Fast-forward to today: ConvNets are everywhere

Classification



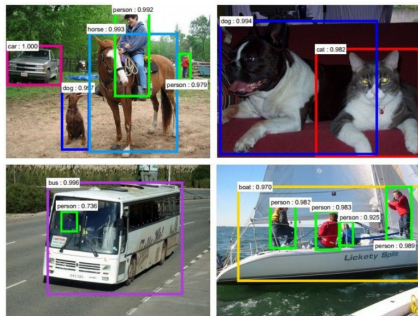
Retrieval



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Fast-forward to today: ConvNets are everywhere

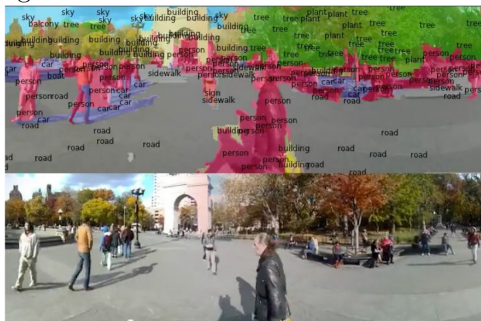
Detection



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[Faster R-CNN: Ren, He, Girshick, Sun 2015]

Segmentation



Figures copyright Clement Farabet, 2012. Reproduced with permission.

[Farabet et al., 2012]

Fast-forward to today: ConvNets are everywhere



self-driving cars

Photo by Lane McIntosh. Copyright CS231n 2017.



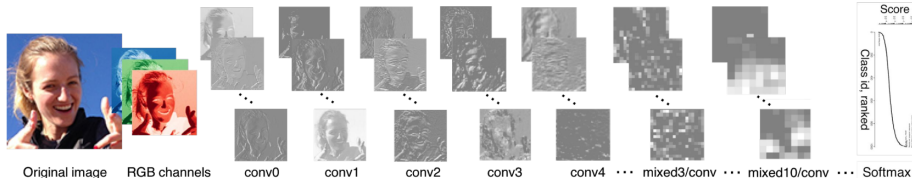
[This image](#) by GBPublic_PR is licensed under [CC-BY 2.0](#)

NVIDIA Tesla line

(these are the GPUs on rye01.stanford.edu)

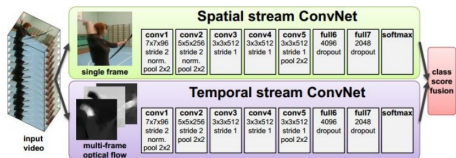
Note that for embedded systems a typical setup would involve NVIDIA Tegras, with integrated GPU and ARM-based CPU cores.

Fast-forward to today: ConvNets are everywhere



[Taigman et al. 2014]

Activations of [inception-v3 architecture](#) [Szegedy et al. 2015] to image of Emma McIntosh, used with permission. Figure and architecture not from Taigman et al. 2014.



[Simonyan et al. 2014]

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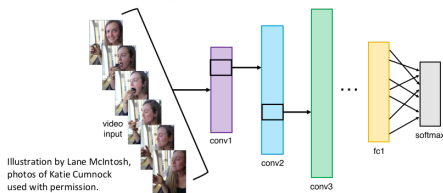


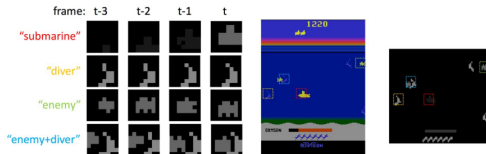
Illustration by Lane McIntosh,
photos of Katie Cumnock
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Fast-forward to today: ConvNets are everywhere

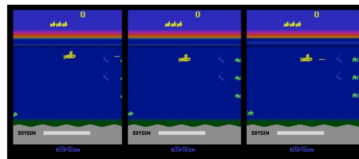


Images are examples of pose estimation, not actually from Toshev & Szegedy 2014. Copyright Lane McIntosh.

[Toshev, Szegedy 2014]

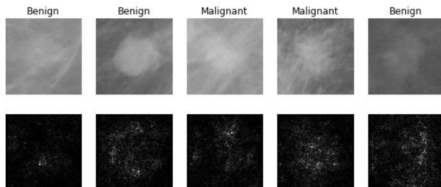


[Guo et al. 2014]



Figures copyright Xiaoxiao Guo, Satinder Singh, Honglak Lee, Richard Lewis, and Xiaoshi Wang, 2014. Reproduced with permission.

Fast-forward to today: ConvNets are everywhere



[Levy et al. 2016]

Figure copyright Levy et al. 2016.
Reproduced with permission.



[Dieleman et al. 2014]

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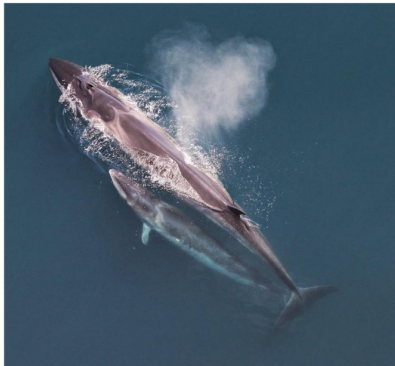
[Sermanet et al. 2011]

[Ciresan et al.]

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Fast-forward to today: ConvNets are everywhere

[This image](#) by Christin Khan is in the public domain and originally came from the U.S. NOAA.



Whale recognition, Kaggle Challenge

Photo and figure by Lane McIntosh; not actual example from Mnih and Hinton, 2010 paper.



Mnih and Hinton, 2010

Fast-forward to today: ConvNets are everywhere

No errors



A white teddy bear sitting in the grass

Minor errors



A man in a baseball uniform throwing a ball

Somewhat related



A woman is holding a cat in her hand

Image Captioning

[Vinyals et al., 2015]
[Karpathy and Fei-Fei, 2015]



A man riding a wave on top of a surfboard



A cat sitting on a suitcase on the floor

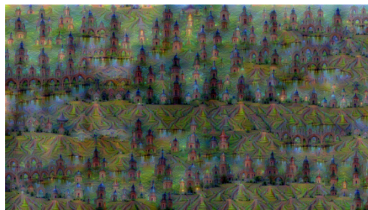
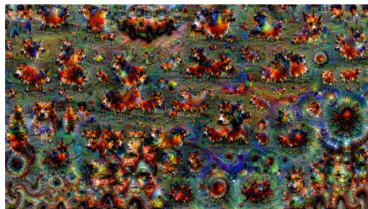


A woman standing on a beach holding a surfboard

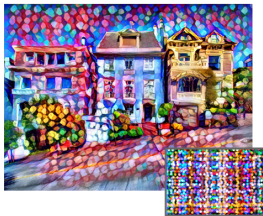
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Captions generated by Justin Johnson using [NeuralTalk2](#)

Fast-forward to today: ConvNets are everywhere



Figures copyright Justin Johnson, 2015. Reproduced with permission. Generated using the Inceptionism approach from a [Google post](#) by Google Research.



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Stylized images copyright Justin Johnson, 2017;
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Gatys et al, "Image Style Transfer using Convolutional Neural Networks", CVPR 2016
Gatys et al, "Controlling Perceptual Factors in Neural Style Transfer", CVPR 2017

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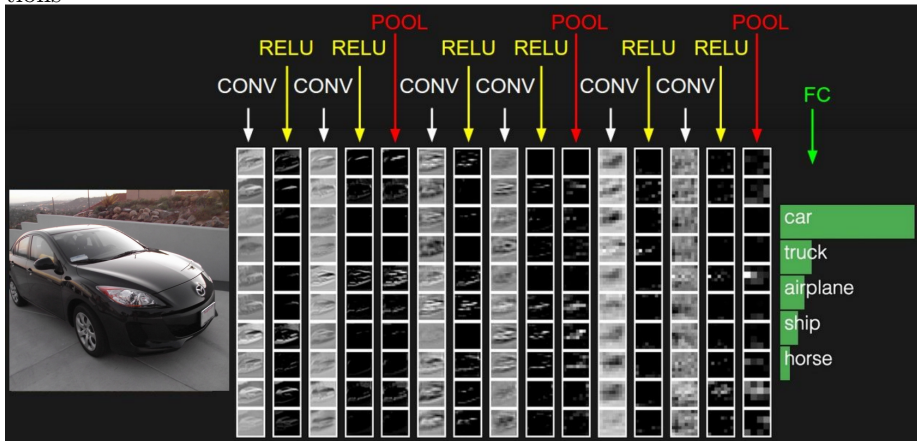
1 History of CNN

2 Convolutional Neural Networks

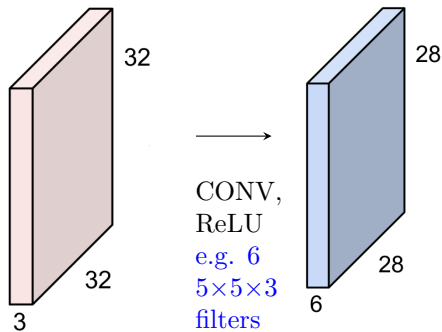
- Preview
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Preview

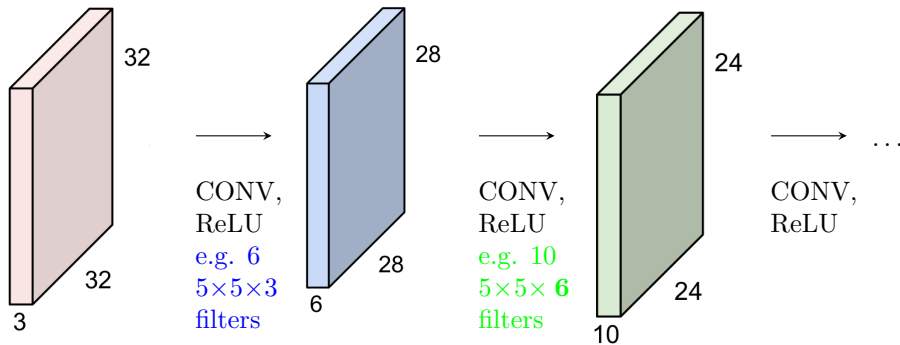
ConvNet is a sequence of Convolution Layers, interspersed with activation functions

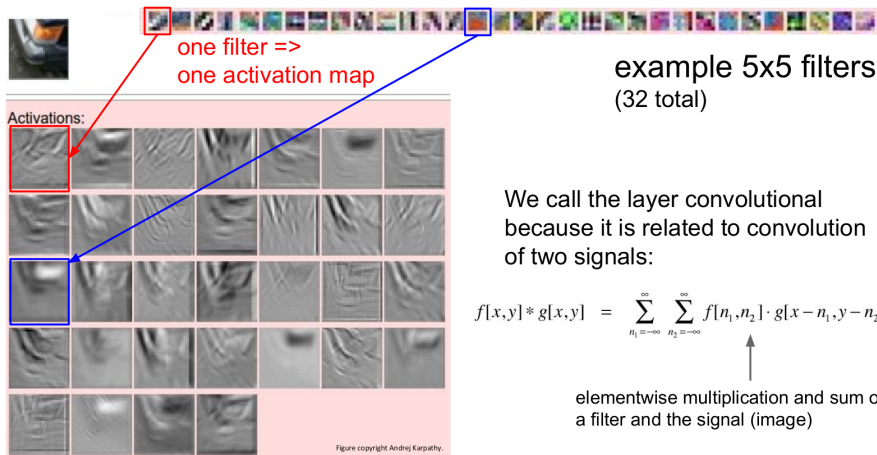


Preview



Preview





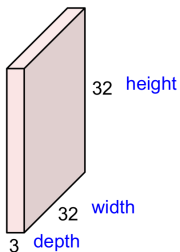
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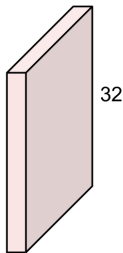
Convolution Layer

$32 \times 32 \times 3$ image $\rightarrow$ preserve spatial structure



Convolution Layer

$32 \times 32 \times 3$ image



$5 \times 5 \times 3$ filter

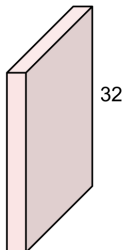


Convolve the filter with the image i.e. “slide over the image spatially, computing dot products”

Convolution Layer

Filters always extend the full depth of the input volume

$32 \times 32 \times 3$ image

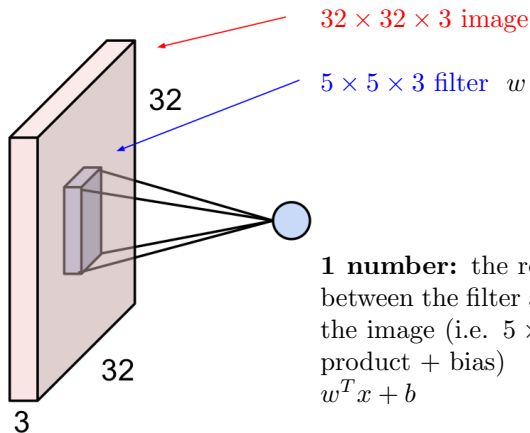


$5 \times 5 \times 3$ filter

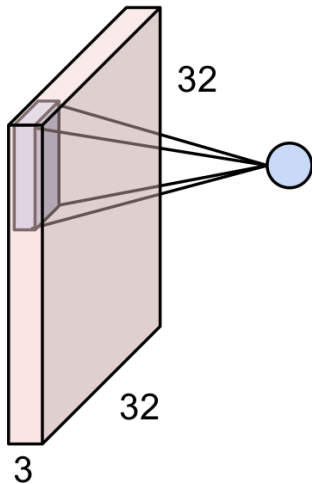


Convolve the filter with the image i.e. “slide over the image spatially, computing dot products”

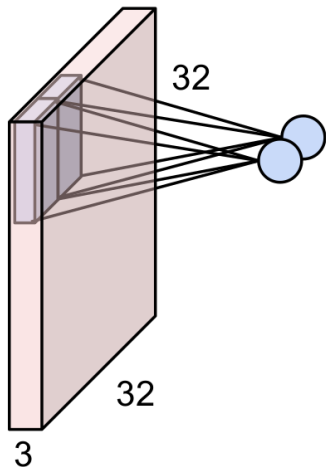
Convolution Layer



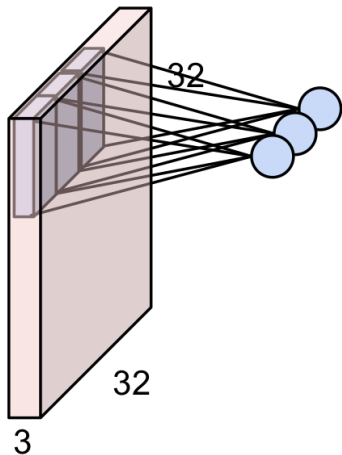
Convolution Layer



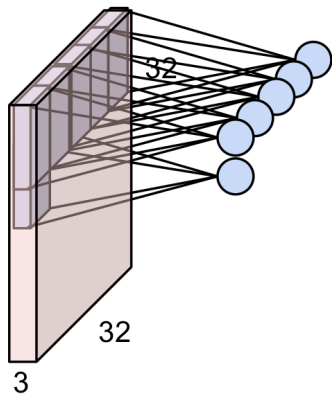
Convolution Layer



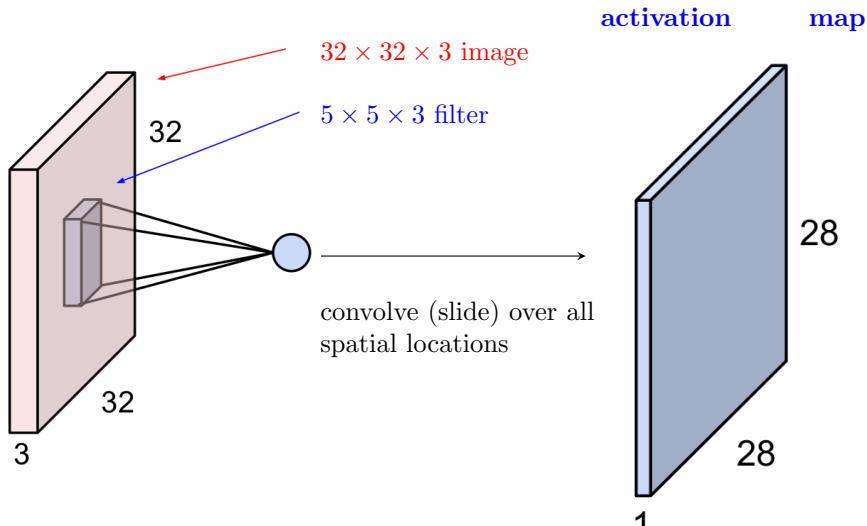
Convolution Layer



Convolution Layer

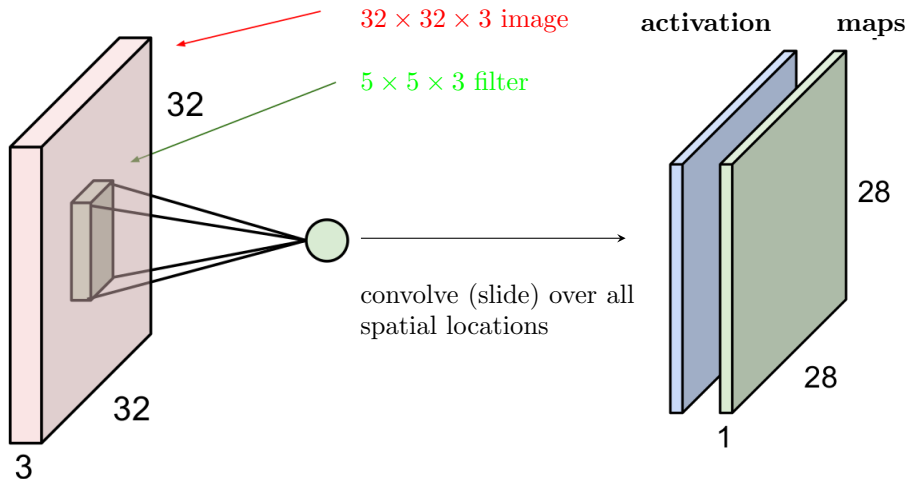


Convolution Layer



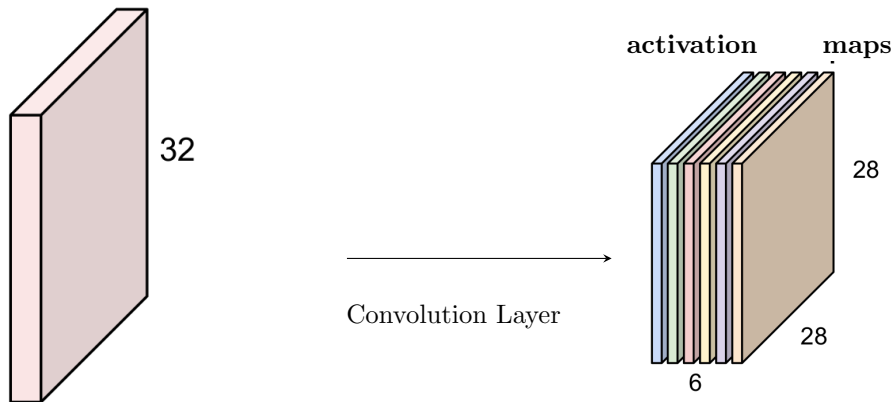
Convolution Layer

consider a second, **green** filter



Convolution Layer

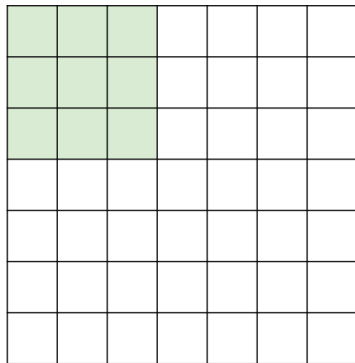
For example, if we had 6 5×5 filters, we'll get 6 separate activation maps:



We stack these up to get a “new image” of size $28 \times 28 \times 6$!

A closer look at spatial dimensions:

7

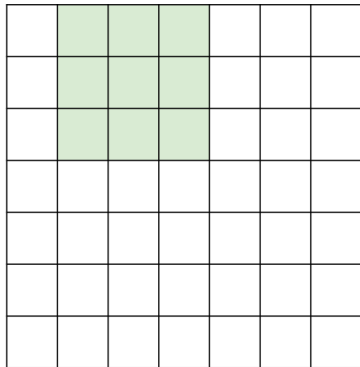


7

7×7 input (spatially) assume 3×3 filter

A closer look at spatial dimensions:

7

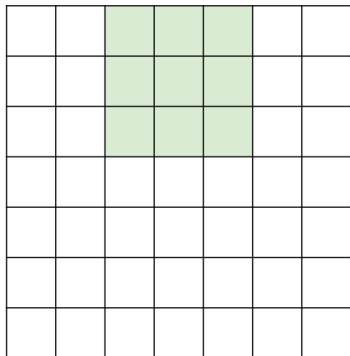


7

7×7 input (spatially) assume 3×3 filter

A closer look at spatial dimensions:

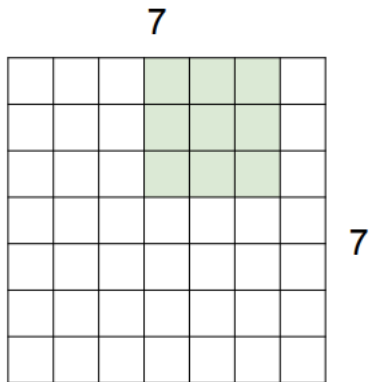
7



7

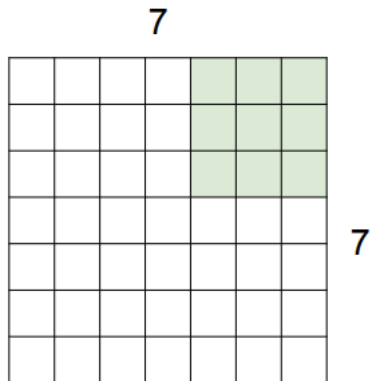
7×7 input (spatially) assume 3×3 filter

A closer look at spatial dimensions:



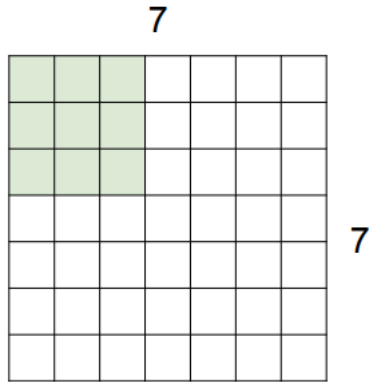
7×7 input (spatially) assume 3×3 filter

A closer look at spatial dimensions:



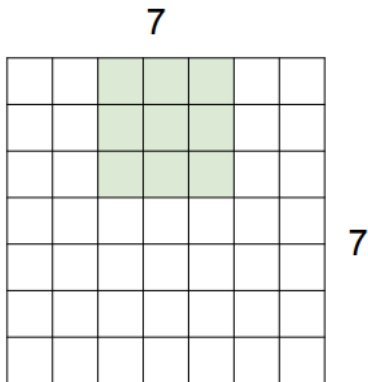
7×7 input (spatially) assume 3×3 filter $\Rightarrow 5 \times 5$ **output**

A closer look at spatial dimensions:



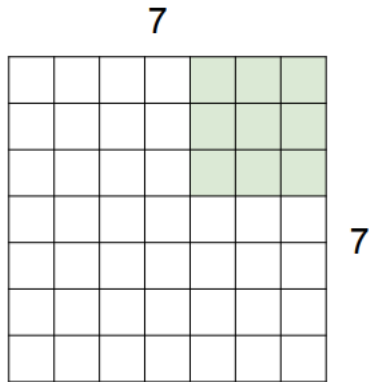
7×7 input (spatially) assume 3×3 filter applied with **stride 2**

A closer look at spatial dimensions:



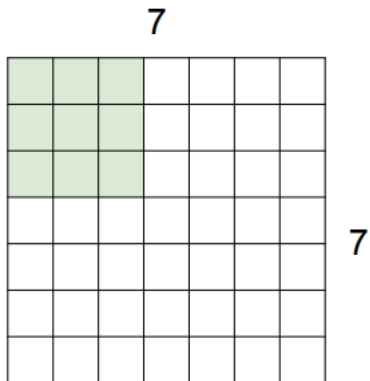
7×7 input (spatially) assume 3×3 filter applied with **stride 2**

A closer look at spatial dimensions:



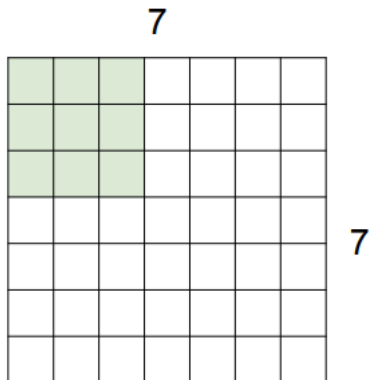
7×7 input (spatially) assume 3×3 filter applied **with stride 2** $\Rightarrow 3 \times 3$ output!

A closer look at spatial dimensions:



7×7 input (spatially) assume 3×3 filter applied **with stride 3** ?

A closer look at spatial dimensions:

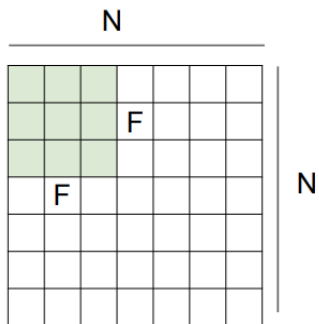


7×7 input (spatially) assume 3×3 filter applied with stride 3?

doesn't fit!

cannot apply 3×3 filter on 7×7 input with stride 3 .

A closer look at spatial dimensions:



Output size: $(N - F) / \text{stride} + 1$ e.g. $N = 7, F = 3$:

stride 1 $\Rightarrow (7 - 3) / 1 + 1 = 5$

stride 2 $\Rightarrow (7 - 3) / 2 + 1 = 3$

stride 3 $\Rightarrow (7 - 3) / 3 + 1 = 2.33$

In practice: Common to zero pad the border

0	0	0	0	0	0			
0								
0								
0								
0								

e.g. input 7×7

3×3 filter, applied with stride 1

pad with 1 pixel border $\Rightarrow$ what is the output?

(recall:)

$$(N - F) / \text{stride} + 1$$

In practice: Common to zero pad the border

0	0	0	0	0	0		
0							
0							
0							
0							

e.g. input 7×7

3x3 filter, applied with **stride 1**

pad with 1 pixel border => what is the output?

7×7 output!

(recall:)

$$(N + 2P - F) / \text{stride} + 1$$

In practice: Common to zero pad the border

0	0	0	0	0	0				
0									
0									
0									
0									

e.g. input 7×7

3x3 filter, applied with **stride 1**

pad with 1 pixel border => what is the output?

7x7 output!

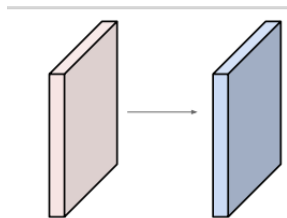
In general, common to see CONV layers with stride 1, filters of size $F \times F$, and zero-padding with $(F - 1)/2$. (will preserve size spatially)

e.g. $F = 3 \Rightarrow$ zero pad with 1

$F = 5 \Rightarrow$ zero pad with 2

$F = 7 \Rightarrow$ zero pad with 3

Examples time:

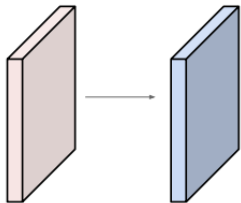


Input volume: $32 \times 32 \times 3$

$10 \times 5 \times 5 \times 3$ filters with stride 1, pad 2

Output volume size: ?

Examples time:



Input volume: $32 \times 32 \times 3$

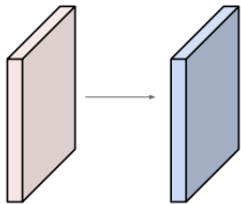
$10 \times 5 \times 5 \times 3$ filters with stride 1, pad 2

Output volume size:

$$(32 + 2 * 2 - 5)/1 + 1 = 32 \text{ spatially, so}$$

$$32 \times 32 \times 10$$

Examples time:

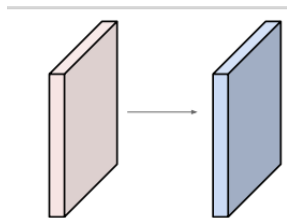


Input volume: $32 \times 32 \times 3$

10 $5 \times 5 \times 3$ filters with stride 1 , pad 2

Number of parameters in this layer?

Examples time:



Input volume: $32 \times 32 \times 3$ $10 \ 5 \times 5 \times 3$ filters with stride 1, pad 2

Number of parameters in this layer? each filter has $5 * 5 * 3 + 1 = 76$ params
(+1 for bias)

$$\Rightarrow 76 * 10 = 760$$

Convolution layer: summary

Let's assume input is $W_1 \times H_1 \times C$ Conv layer needs 4 hyperparameters:

- Number of filters K
- The filter size F
- The stride S
- The zero padding P

This will produce an output of $W_2 \times H_2 \times K$ where:

$$- W_2 = (W_1 - F + 2P) / S + 1$$

$$- H_2 = (H_1 - F + 2P) / S + 1$$

Number of parameters: F^2CK and K biases

Convolution layer: summary

Let's assume input is $W_1 \times H_1 \times C$ Conv layer needs 4 hyperparameters:

- Number of filters K
- The filter size F
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- The zero padding P

This will produce an output of $W_2 \times H_2 \times K$ where:

$$- W_2 = (W_1 - F + 2P) / S + 1$$

$$- H_2 = (H_1 - F + 2P) / S + 1$$

Number of parameters: F^2CK and K biases

Common settings:

$$K = (\text{powers of 2, e.g. } 32, 64, 128, 512)$$

$$- F = 3, S = 1, P = 1$$

$$- F = 5, S = 1, P = 2$$

$$- F = 5, S = 2, P = ? \text{ (whatever fits)}$$

$$- F = 1, S = 1, P = 0$$

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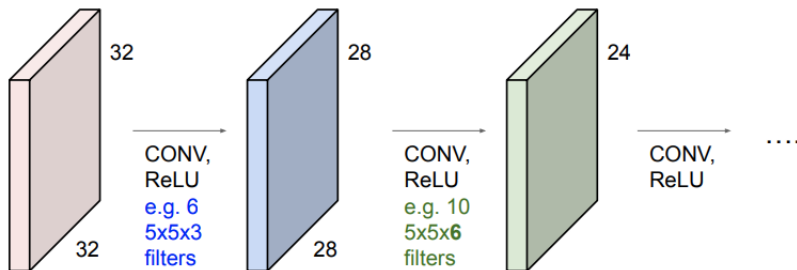
- Preview
- Convolutional layer
- Pooling layer
- Fully connected layer

Pooling layer

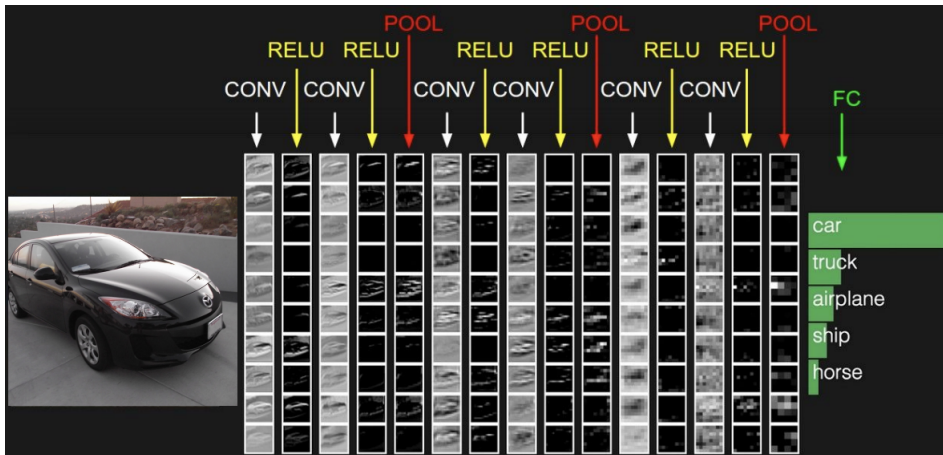
Remember back to...

E.g. 32×32 input convolved repeatedly with 5×5 filters shrinks volumes spatially! ($32 \rightarrow 28 \rightarrow 24 \dots$).

Shrinking too slow is very costly; Shrinking too fast is not good, doesn't work well.

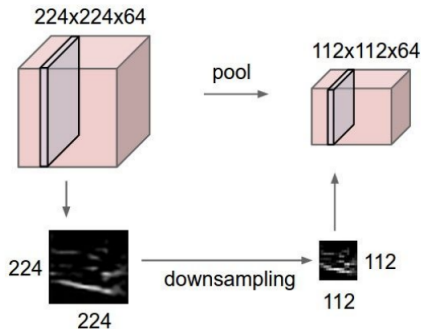


Pooling layer

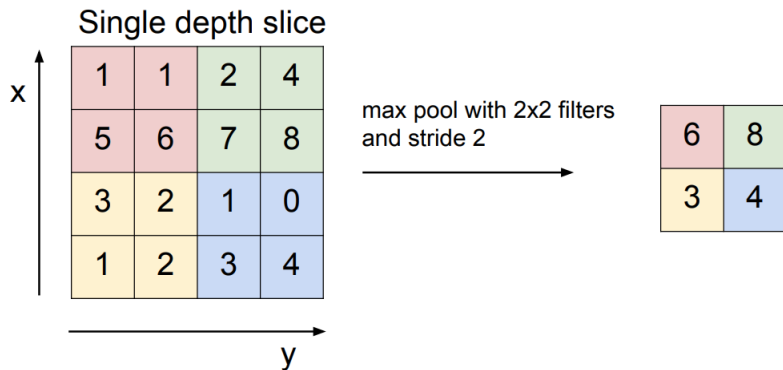


Pooling Layer

- makes the representations smaller and more manageable
- operates over each activation map independently:



MAX POOLING



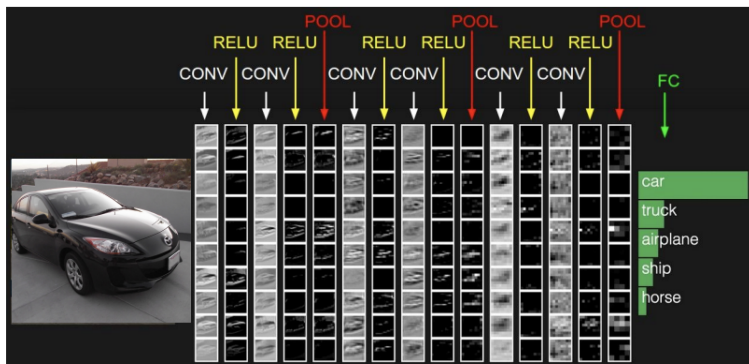
1 History of CNN

2 Convolutional Neural Networks

- Preview
- Convolutional layer
- Pooling layer
- Fully connected layer

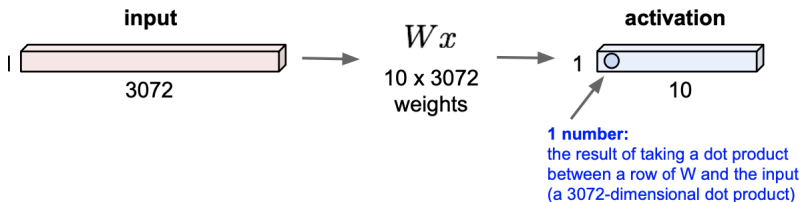
Fully Connected Layer (FC layer)

Contains neurons that connect to the entire input volume, as in ordinary Neural Networks



Fully Connected Layer (FC layer)

32x32x3 image -> stretch to 3072 x 1



Further reading

Some resources for learning CNNs:

- Course: CS231n: Deep Learning for Computer Vision
- Packages: Pytorch, Tensorflow